



No more 16-cylinder by Bugatti

Bugatti CEO says Chiron will be the last 16-cylinder engine car built by the company.
<https://digit.in/oct18-91-1>



Streaming everywhere

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on Earth, that do not need oxygen to survive. In our investigation, we wanted to use a more holistic approach to finding biosignatures, and not use life as we know it on Earth as the only basis. If we are looking for aliens, we have to take a broader approach."

The atmospheric chemical disequilibrium approach does not make any assumptions about the biochemistry of the life forms on a planet. Instead, it studies the gases in the atmosphere of the planet, and checks if these gases could be there if they were not being constantly generated as the byproducts of the biochemistry of any potential life forms. On the Earth, if all of life were to suddenly disappear, then the atmosphere would change to a different chemical composition over a period of time. While all the planets in the solar system show some kind of disequilibrium, the Earth has among the most. The Saturnian moon Titan has a larger disequilibrium than even the Earth, keeping the possibility open that we might find extraterrestrial life closer back home.

During the course of the study, the authors eliminated every single natural process that could explain the large amounts of methane that was observed in the atmospheres of Trappist-1 e and f. Some of the natural sources of methane include volcanoes and natural chemical reactions in wetlands. The Trappist-1 e and f planets show a significant thermal disequilibrium as compared to other planets in the system, and this cannot be explained by any known geological processes. If unknown factors are discarded, life forms are the only remaining explanation of the methane found on Trappist-1 e and f. The next highest disequilibrium in the system is that of Trappist-1 g.

Lucas Guerrero, co author of the study says, "The Trappist-1 system is the most heavily studied system, after our own solar system. We know more about these worlds than any other exoplanet system. Right from its discovery, there was this sense of awe and wonder about a promising system where there were just so many reasons why life could exist. Now, we have found even more reasons."

The new finding bolsters evidence gathered by other telescopes that there is life in the Trappist-1 system. Follow on observation campaigns are being planned by a number of ground and space based astronomical instruments. The ARIEL (Atmospheric Remote-sensing Exoplanet Large-survey) to be launched by the European Space Agency next year is expected to play an important role in further studies of the Trappist-1 system.

The James Webb Space Telescope is the most powerful space telescope ever built. Webb is an international collaboration led by NASA and its partners, the European Space Agency (ESA) and the Canadian Space Agency (CSA). The telescope is operated by the Space Telescope Science Institute (STScI), located at the Homewood campus of Johns Hopkins University. 

Credit: ESO/M. Kornmesser